

CarbonCoin - Carbon Credit Marketplace Platform

Product Requirements Document (PRD)

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Executive Summary

CarbonCoin is a blockchain-inspired carbon credit marketplace that democratizes carbon credit participation by allowing individuals, small landowners, farmers, and local communities to monetize their environmental contributions. The platform leverages AI-powered verification and satellite imagery to validate green assets like trees, land, and plantations, converting them into tradable carbon credits called “Carbon Coins.”

Key Value Propositions

- **For Individuals/Farmers:** Turn environmental efforts into measurable, monetizable value
 - **For Companies:** Access verified, scalable carbon credits for sustainability goals
 - **For Society:** Promote inclusive, transparent, and impactful carbon markets
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Product Overview

Mission Statement

To democratize the carbon credit economy by making it accessible to everyday environmental contributors while maintaining verification integrity through advanced technology.

Vision

A world where every positive environmental action is recognized, measured, and rewarded, contributing to global climate goals through an inclusive and transparent marketplace.

Core Concept

CarbonCoin enables users to register green assets (trees, land, plantations), submit verification details using AI and satellite technology, and earn Carbon Coins (digital carbon credits) that can be traded on the marketplace.

Problem Statement

Traditional carbon credit markets are dominated by large institutions, excluding small-scale environmental contributors. Manual verification processes are slow, unreliable, and prone to fraud, limiting trust and scalability.

Solution

An AI-powered, satellite-verified carbon credit marketplace that: - Democratizes access to carbon markets - Uses technology for fast, accurate verification - Creates transparent, trustworthy transactions - Enables monetization of environmental contributions

User Personas

Persona 1: Individual Environmental Contributor (Farmer/Landowner)

- **Name:** Ramesh Patel
- **Role:** Small-scale farmer
- **Goals:** Monetize tree plantation efforts, contribute to environmental causes
- **Pain Points:** Lack of access to carbon markets, complex verification processes
- **Technology Usage:** Basic smartphone user, familiar with simple apps

Persona 2: Corporate Sustainability Manager

- **Name:** Priya Sharma
- **Role:** Sustainability manager at a corporation
- **Goals:** Meet corporate sustainability targets, purchase verified carbon credits
- **Pain Points:** Difficulty finding reliable, verified carbon credits
- **Technology Usage:** Advanced user, comfortable with business platforms

Persona 3: Environmental Organization

- **Name:** Green Earth Foundation
 - **Role:** NGO focused on environmental conservation
 - **Goals:** Scale environmental impact, support local communities
 - **Pain Points:** Limited funding, difficulty proving impact
 - **Technology Usage:** Moderate, prefers user-friendly platforms
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Features & Functionalities

1. User Registration & Authentication

- **Registration:** Email, password, name, phone, address, user type (seller/company)
- **Login:** Secure JWT-based authentication
- **Profile Management:** Update personal information
- **Logout:** Session termination

2. Identity Verification (KYC)

- **Document Upload:** Selfie and Aadhar card for sellers
- **Face Verification:** AI-powered face matching between selfie and Aadhar photo
- **Status Tracking:** Verification status updates

3. Property Management

- **Property Registration:** Title, description, geotagged images, address, boundary coordinates, area
- **Image Upload:** Multiple property images with geotagging
- **Property Listing:** View all registered properties
- **My Properties:** Manage owned properties
- **Property Updates:** Modify property details
- **Property Deletion:** Remove properties

4. Carbon Coin System

- **Wallet Management:** Track Carbon Coin balance
- **Transaction History:** View all coin transactions
- **Coin Transfer:** Send coins to other users (seller to company)
- **Leaderboard:** Top contributors display
- **Marketplace Stats:** Overall platform statistics

5. Company Features

- **Company Registration:** Business account creation

- **Document Upload:** Company logo and registration documents
- **Profile Management:** Update company information
- **View Companies:** Browse all registered companies

6. Admin/Superuser Features

- **User Management:** View all users
 - **Property Oversight:** Monitor all properties
 - **Verification Management:** Approve/reject verifications
 - **Analytics Dashboard:** Platform performance metrics
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Technical Architecture

Backend Stack

- **Language:** JavaScript (Node.js)
- **Framework:** Express.js
- **Database:** MongoDB (with Mongoose ODM)
- **Authentication:** JWT (JSON Web Tokens)
- **File Upload:** Multer with Cloudinary integration
- **Logging:** Morgan
- **Security:** Bcrypt for password hashing

Frontend Stack

- **Framework:** Next.js 16.1.3
- **Language:** React 19.2.3
- **Styling:** Tailwind CSS
- **Runtime:** Client-side rendering

KYC Service

- **Language:** Python
- **Framework:** Flask
- **Face Recognition:** DeepFace library
- **Database:** MongoDB integration
- **Image Processing:** OpenCV, RetinaFace

Third-party Integrations

- **Cloud Storage:** Cloudinary for image hosting
- **Database:** MongoDB Atlas or local MongoDB
- **Face Verification:** DeepFace for biometric verification

Data Models

User Model

- email (String, required, unique)
- password (String, required)
- name (String, required)
- phone (String, required, unique)
- address (String)
- verified (Boolean, default: false)
- type (String, enum: ["seller", "company"], required)

Seller Model

- userId (ObjectId, ref: "User", required, unique)
- selfieUrl (String)
- aadharUrl (String)

Company Model

- userId (ObjectId, ref: "User", required, unique)
- logoUrl (String)
- registrationDocUrl (String)

Property Model

- SellerId (ObjectId, ref: "User", required)
- title (String, required)
- description (String, required)
- geotaggedImageUrls (Array of Strings, required)
- address (String, required)
- boundaryCoordinates (Array of objects with lat/lng, required)
- areaInSqFt (Number, required)

CarbonCoin Model

- ownerId (ObjectId, ref: "User", required)
- amount (Number, required)
- history (Array of transaction objects with date, amount, propertyId)

API Endpoints

Authentication Endpoints (/api/auth)

- POST /register - User registration
- POST /login - User authentication
- GET /profile - Get user profile (protected)

- PUT /profile - Update user profile (protected)
- POST /logout - User logout (protected)

Seller Endpoints (/api/sellers)

- POST /upload-documents - Upload verification documents (selfie + Aadhar) (protected, seller only)
- GET /profile - Get seller profile (protected, seller only)
- GET /all - Get all sellers (protected)

Property Endpoints (/api/properties)

- POST / - Create property (protected, seller only, verified)
- GET / - Get all properties (public)
- GET /my-properties - Get seller's properties (protected, seller only)
- GET /:id - Get single property
- PUT /:id - Update property (protected, seller only)
- DELETE /:id - Delete property (protected, seller only)

Carbon Coin Endpoints (/api/carboncoins)

- GET /balance - Get coin balance (protected)
- GET /history - Get transaction history (protected)
- POST /transfer - Transfer coins (protected, verified user)
- GET /leaderboard - Get top contributors (public)
- GET /stats - Get marketplace statistics (public)

Company Endpoints (/api/companies)

- POST /upload-documents - Upload company documents (logo + registration) (protected, company only)
- GET /profile - Get company profile (protected, company only)
- GET /all - Get all companies (protected)

KYC Service Endpoints (http://localhost:5000)

- POST /api/verify-face - Face verification between Aadhar and selfie
- GET /api/health - Health check
- GET / - API documentation

User Interface

Frontend Pages

1. **Homepage (/)**
 - Hero section with value proposition

- About section explaining the platform
- Feature highlights
- 2. **Sign Up (/signup)**
 - User registration form
 - Account type selection (seller/company)
- 3. **Login (/login)**
 - User authentication form
- 4. **Dashboard (/dashboard)**
 - User profile overview
 - Property listings
 - Carbon coin balance
 - Recent activity
- 5. **Property Verification (/property-verification)**
 - Form to register new properties
 - Image upload functionality
 - Geotagging interface
- 6. **Identity Verification (/verify-identity)**
 - Document upload interface
 - Verification status tracking
- 7. **About (/about)**
 - Detailed platform information
 - Mission and vision statements

UI Components

- **Navbar:** Navigation across the platform
 - **Footer:** Contact information and legal links
 - **Forms:** For registration, property submission, and document uploads
 - **Cards:** For displaying properties and user profiles
 - **Charts/Graphs:** For displaying marketplace statistics
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Security & Compliance

Authentication & Authorization

- JWT-based session management
- Role-based access control (seller/company/admin)
- Password hashing with bcrypt
- Protected routes middleware

Data Protection

- Secure file uploads with validation
- Encrypted storage of sensitive information
- Input sanitization and validation

- Rate limiting for API endpoints

KYC & Verification

- Biometric face verification
- Document authenticity checks
- Secure storage of personal documents
- Privacy-compliant data handling

Compliance Considerations

- GDPR compliance for data protection
 - Local regulations for financial transactions
 - Carbon credit certification standards
 - Environmental verification protocols
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Development Timeline

Phase 1: MVP (Weeks 1-4)

- User authentication system
- Basic property registration
- Simple carbon coin tracking
- Frontend homepage and basic UI

Phase 2: Verification (Weeks 5-8)

- KYC service integration
- AI-powered verification system
- Property validation workflow
- Enhanced UI/UX

Phase 3: Marketplace (Weeks 9-12)

- Carbon coin transfer functionality
- Company onboarding
- Transaction history
- Leaderboard and analytics

Phase 4: Scaling (Weeks 13-16)

- Advanced verification features
 - Mobile app development
 - API documentation
 - Performance optimization
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Success Metrics

User Adoption

- Number of registered users
- Daily/Monthly active users
- User retention rate
- Conversion from registration to first property submission

Platform Activity

- Number of verified properties
- Total carbon coins issued
- Number of successful transactions
- Average transaction value

Business Impact

- Carbon credits generated
- Environmental impact measurement
- Company partnerships
- Revenue from transaction fees

Technical Performance

- API response times
 - System uptime
 - Error rates
 - User satisfaction scores
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Future Enhancements

Advanced Features

- Blockchain integration for transparency
- IoT sensor integration for real-time monitoring
- Advanced analytics and reporting
- Mobile applications (iOS/Android)

Market Expansion

- International carbon credit standards
- Integration with existing carbon markets
- Corporate sustainability reporting tools
- Partnership with environmental organizations

Technology Improvements

- Machine learning for automated verification
 - AR/VR for property visualization
 - Predictive analytics for carbon credit valuation
 - Automated compliance checking
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Risk Assessment

Technical Risks

- Scalability challenges with growing user base
- Accuracy of AI verification systems
- Security vulnerabilities in financial transactions
- Third-party service dependencies

Market Risks

- Regulatory changes in carbon credit markets
- Competition from established players
- Economic factors affecting carbon credit prices
- Adoption challenges in traditional markets

Mitigation Strategies

- Regular security audits and penetration testing
 - Continuous monitoring and improvement of AI models
 - Legal consultation for regulatory compliance
 - Diversified revenue streams and partnerships
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Conclusion

CarbonCoin represents a significant opportunity to democratize the carbon credit market while leveraging technology to ensure verification integrity. The platform addresses key pain points in the current carbon credit ecosystem by making participation accessible to individuals and small organizations while maintaining the trust and transparency required for a functioning marketplace.

The success of CarbonCoin depends on achieving critical mass of both contributors and buyers, establishing trust through robust verification mechanisms, and continuously improving the user experience to encourage adoption and retention.